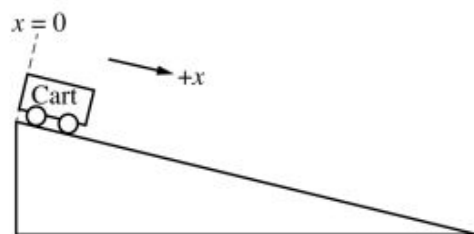


AGENDA FOR 01-04-24A/5B



1. WARMUPS (GRAVITATION)
2. SELF ASSESSMENT
3. SEMESTER 1 REVIEW
4. ORBITAL SPEED
5. KEPLER'S LAWS
6. **GRAVITATION QUIZ NEXT CLASS!**



2. (12 points, suggested time 25 minutes)

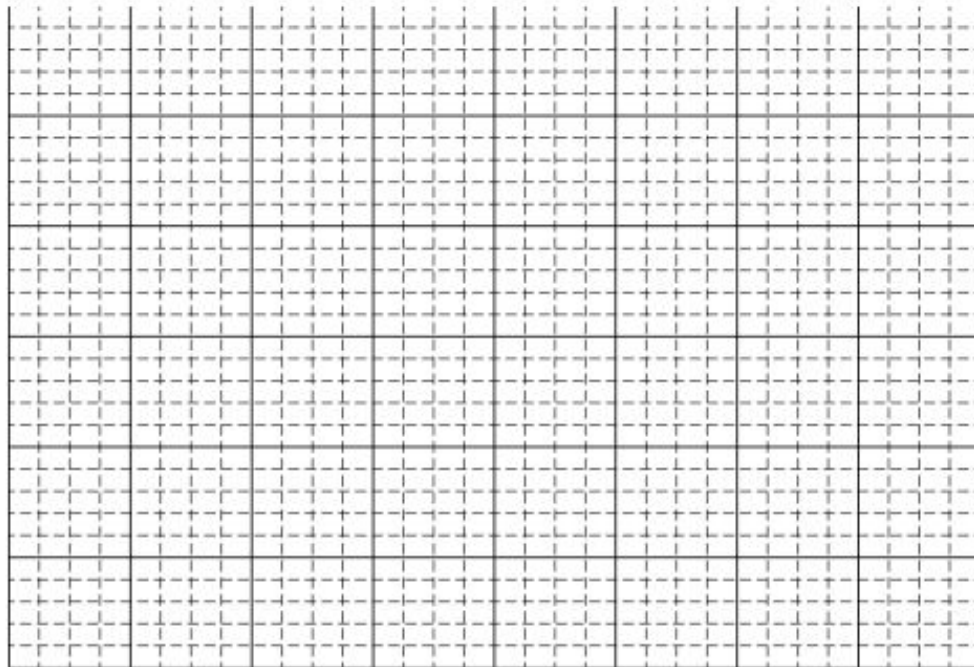
- (a) Students conduct an experiment to determine the acceleration a of a cart. The cart is released from rest at the top of the ramp at time $t = 0$ and moves down the ramp. The x -axis is defined to be parallel to the ramp with its origin at the top, as shown in the figure. The students collect the data shown in the following table.

	Position x (m)	Time t (s)	
	0.06	0.39	
	0.14	0.59	
	0.24	0.77	
	0.37	0.96	
	0.55	1.20	

- i. Indicate which quantities could be graphed to yield a straight line whose slope could be used to determine the acceleration a of the cart. You may use the remaining columns in the table, as needed, to record any quantities (including units) that are not already in the table.

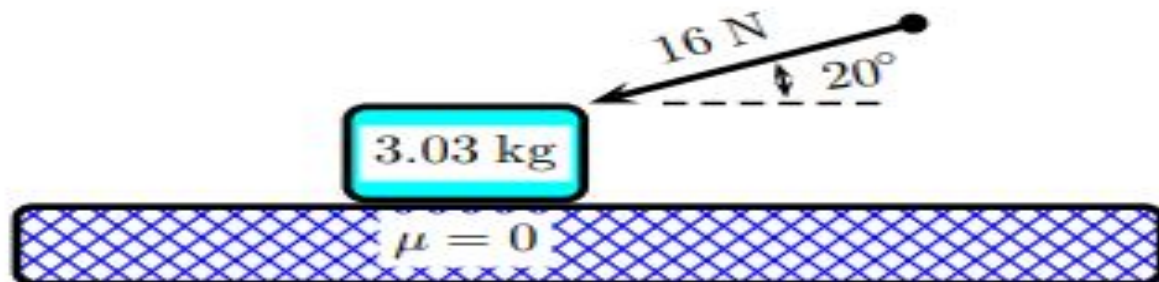
Vertical axis: _____ Horizontal axis: _____

- ii. On the following grid, plot the appropriate quantities to create a graph that can be used to determine the acceleration a of the cart as it rolls down the ramp. Clearly scale and label all axes (including units), as appropriate. Draw a straight line that best represents the data.



- iii. Using the line you drew in part (a)(ii), calculate an experimental value for the acceleration a of the cart as it rolls down the ramp.

A block of mass 3.03 kg is pushed 3.1 m along a frictionless horizontal table by a constant 16 N force directed 20° below the horizontal.



Find the work done by the applied force.
Answer in units of J. Answer in units of J.

002 (part 2 of 4) 10.0 points

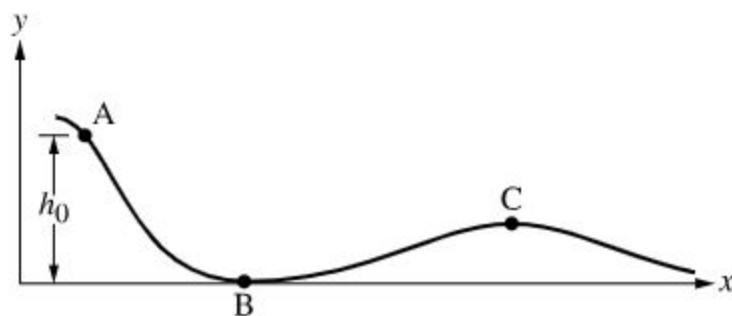
Find the work done by the normal force exerted by the table.

Answer in units of J. Answer in units of J.

003 (part 3 of 4) 10.0 points

Find the work done by the force of gravity.

Answer in units of J. Answer in units of J.



Note: Figure not drawn to scale.

A sled is released from rest at Point A on a track where friction between the sled and the track is negligible. Point A is at a vertical position of $y = h_0$, as shown. The sled remains in contact with the track as the sled passes through points B and C.

Let System **A** consist of the sled only and System **B** consist of the sled and Earth. Which of the following correctly indicates at what point, if any, the mechanical energy E of each system is a maximum?

	System A	System B
A	None, because E is constant	None, because E is constant
B	None, because E is constant	Point A
C	Point B	None, because E is constant
D	Point B	Point A

An object is released from rest near the surface of Earth. Initially, the object accelerates as it begins to fall towards the Earth's surface. Earth's atmosphere exerts a resistive force on the object, and it reaches a terminal velocity before landing. Which of the following claims about the gravitational potential energy of the object-Earth system is true?

A The gravitational potential energy remains constant during the initial acceleration period but then decreases as the object falls at terminal velocity.



B The gravitational potential energy decreases during the initial acceleration period but then remains constant as the object falls at terminal velocity.



C The gravitational potential energy remains constant both during the initial acceleration period and as the object falls at terminal velocity.



D The gravitational potential energy decreases both during the initial acceleration period and as the object falls at terminal velocity.



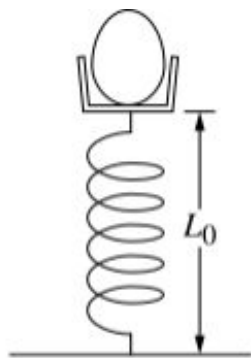


Figure 1

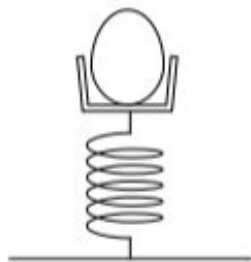
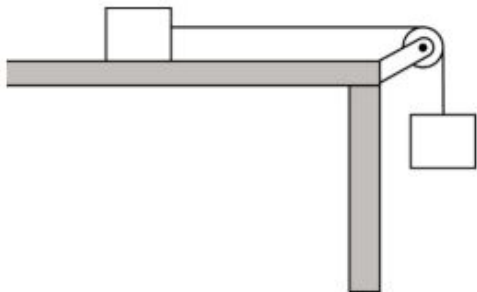


Figure 2

A vertical spring is at its unstretched length L_0 when an egg is gently placed on the spring, as shown in Figure 1. The egg is then slowly lowered until it reaches equilibrium and remains at rest, as shown in Figure 2. Which of the following correctly describes the change in the gravitational potential energy ΔU_g of the Earth-egg system and the change in the potential energy of the spring, ΔU_s ?

	ΔU_g	ΔU_s
A	Positive	Positive
B	Positive	Negative
C	Negative	Positive
D	Negative	Negative



A block is set on a table, where there is negligible friction between the block and the table. The block is connected to an identical hanging block by a lightweight string that passes over an ideal pulley as shown. When the blocks are released from rest, the two-block system gains kinetic energy because work is done on the system. Which type of force or forces make a nonzero contribution to the net work done on the two-block system?

(A) Gravitational force only



(B) Gravitational force and tension only

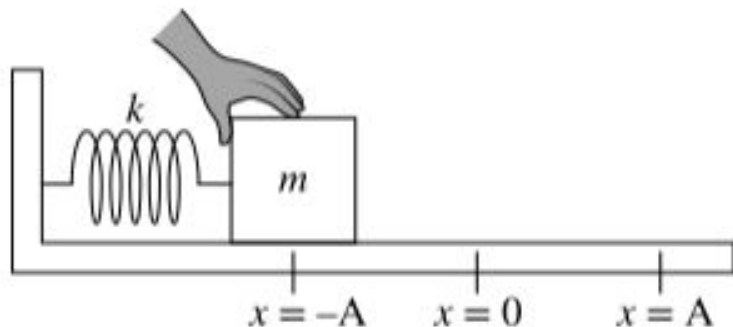


(C) Gravitational force and normal force only



(D) Gravitational force, tension, and normal force

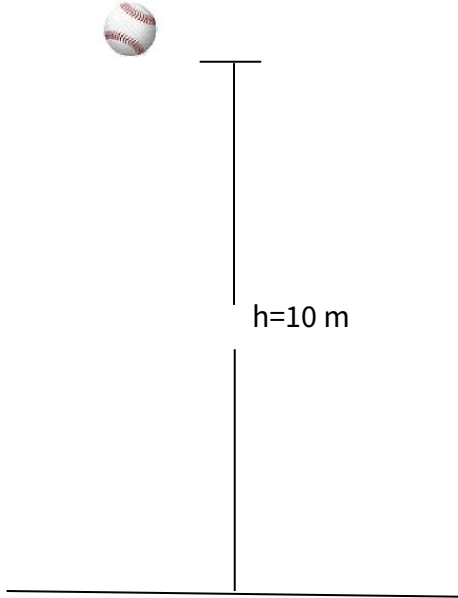


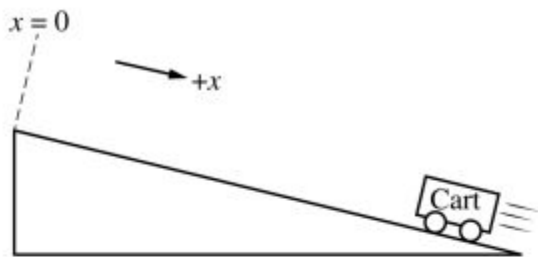


A block is attached to a compressed, horizontal spring. A person is initially holding the block in place at position $x = -A$ as shown in the figure. The person then slowly moves the block at a constant speed through position $x = 0$, at which point the spring is at its relaxed length, and then to position $x = +A$. Which of the following correctly describes how the potential energy in the spring-block system changes as the block is moved from $x = -A$ to $x = +A$?

- (A) The potential energy decreases throughout the motion
- (B) The potential energy increases throughout the motion
- (C) The potential energy decreases and then increases
- (D) The potential energy increases and then decreases

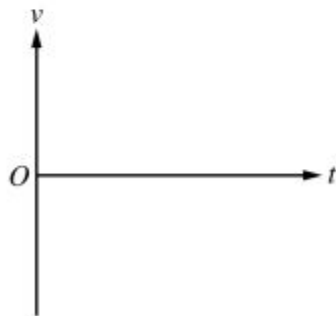
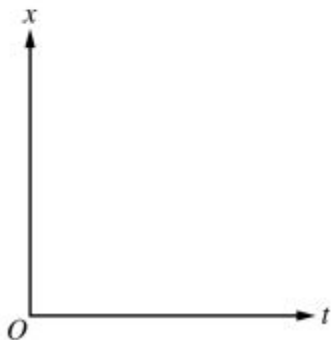
A 0.3 kg baseball is released from rest in the absence of air resistance at a height of 10 m as shown below. Use the work-mechanical energy theorem to determine the speed of the baseball right before it hits the ground. Solve it for the following systems: a) ball only b) ball-earth





The students want to confirm that the acceleration is the same whether the cart rolls up or down the ramp. The students start the cart at the bottom and give the cart a quick push so that it rolls up the ramp and momentarily comes to rest. The x -axis is still defined to be parallel to the ramp with the origin at the top.

- (d) On the following graphs, sketch the position x and velocity v as functions of time t that correspond to the scenario shown while the cart moves up the ramp.



WARMUP

Calculate the gravitational force between a 6.5×10^{27} kg planet and its 2.8×10^{20} kg moon. The center of the planet's moon is 5.2×10^6 m from its center.

WARMUP

Two identical masses separated by a distance of r , between their centers have a gravitational force of 100 N acting on each of them opposite in direction. What is the new force when one mass is tripled and the distance between them is changed to one-fourth of its original value.

WARMUP

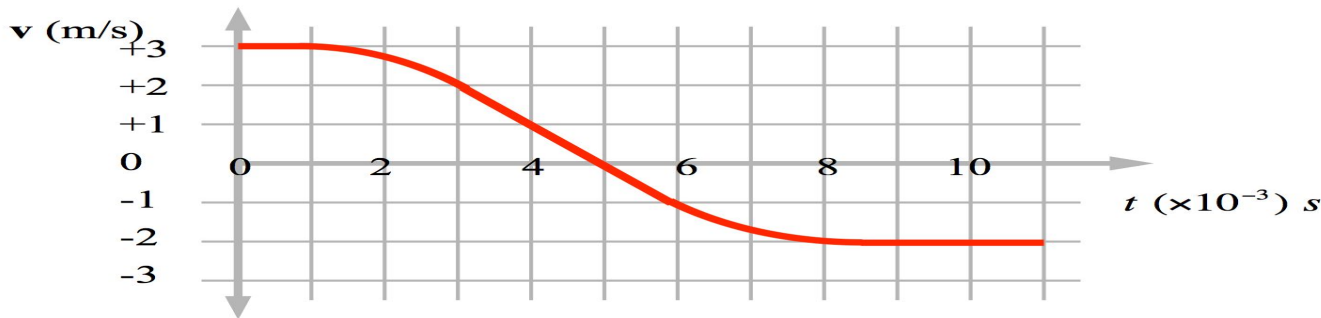
A mass is taken from the surface of the moon to a distance of $4r$ above the surface of the moon. What is the acceleration due to gravity at its new location if the acceleration due to the moon's gravity at its surface is, g_m

WARMUP

Planet green has an acceleration due to gravity 7 times that of planet brown. Planet green's radius is 3 times that of planet brown. What is the ratio of planet green's mass to planet brown's mass?

K-Numbered Head Together 1

Question:

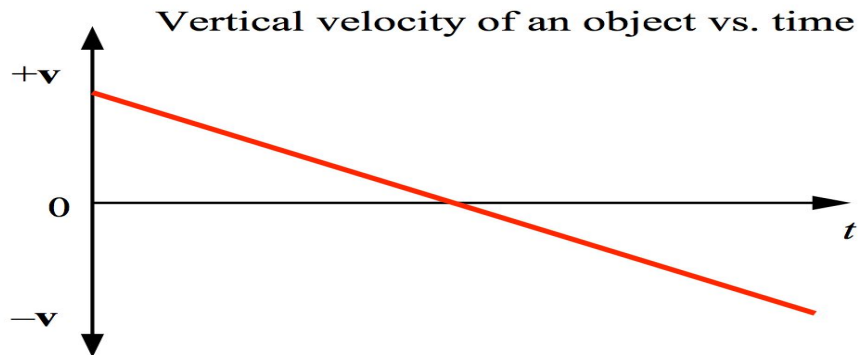


A billiard ball rolling across a table in the $+x$ direction at 3 m/s hits the edge of the table at a perpendicular angle, and bounces back in the $-x$ direction, now traveling at 2 m/s in the opposite direction. The greatest magnitude of acceleration for the billiard ball was at time

- a. $t = 1 \times 10^{-3}$ s
- b. $t = 2 \times 10^{-3}$ s
- c. $t = 5 \times 10^{-3}$ s
- d. $t = 7 \times 10^{-3}$ s
- e. $t = 9 \times 10^{-3}$ s

K-Numbered Head Together 1

Question:



The *vertical velocity vs. time* motion for an object is shown above, with positive velocity indicating a speed in the upward direction. Based on this graph,

- the object *must* be traveling in a negative direction throughout its motion
- the object *must* be traveling with a constant velocity throughout its motion
- the object has an acceleration that is changing
- the object *must* have been thrown straight up into the air
- the vertical motion of the object is consistent with that of a projectile

AGENDA FOR 01-08A/9B



WARMUP

SELF ASSESSMENT

GRAVITATION QUIZ

ENERGY & SYSTEMS

WARMUP

It is observed that Jupiter's largest moon Ganymede is 3 times the mass of a newly discovered planet. If the acceleration due to gravity on Ganymede is 9 times the acceleration due to gravity on the new planet, how does the planet's radius compare to Ganymede's radius?

AGENDA FOR 12-4B/5A



UNITS POP QUIZ in SCHOOLOGY-
starts on the bell (10 minutes)

OPEN SYSTEMS

CLOSED SYSTEMS

Question:

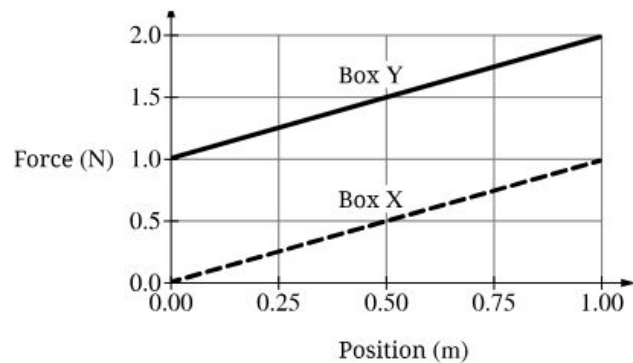
A moon of mass m orbits a planet of mass M in a perfectly circular orbit of radius R , with a force of gravitational attraction between the two bodies of F_g . How much Work is done on the moon by the planet during a single orbit of the moon ?

- a. 0
- b. $F_g R$
- c. $F_g M m R$
- d. $F_g 2\pi R$
- e. $F_g \pi R^2$

Question:

A 10-meter long, vertical cannon is used to accelerate a 1.0-kg ball straight up into the air. A constant force of 13.2-Newtons is used to accelerate the bowling ball up the length of the cannon. What is the ball's approximate velocity as it leaves the cannon (assuming no energy loss to friction)?

- a. 29 m/s
- b. 16 m/s
- c. 14 m/s
- d. 9 m/s
- e. 8 m/s



Boxes **X** and **Y** are pushed horizontally the same distance by forces with varying magnitudes. The magnitudes of the forces exerted on each box as a function of their positions are shown in the graph. What is the ratio $W_X : W_Y$ of the work done on each box by the forces?

(A) 1 : 1

(B) 1 : 2

(C) 1 : 3

(D) 1 : 4

AGENDA FOR 01-17A/18B



WARMUP

WORK-ENERGY

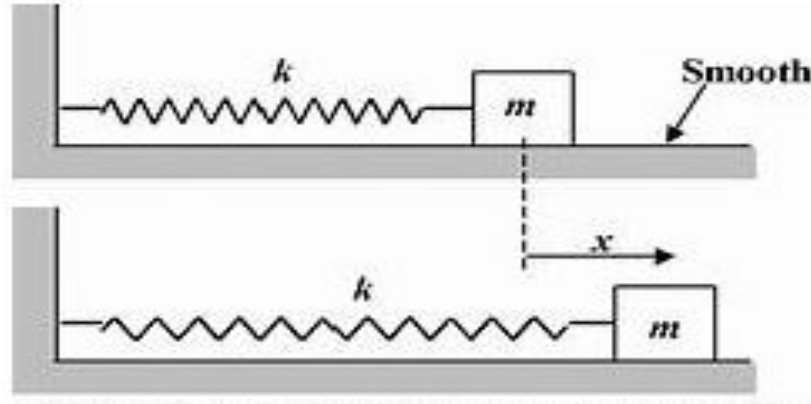
AGENDA FOR 01-18-2022/01-19 -2022

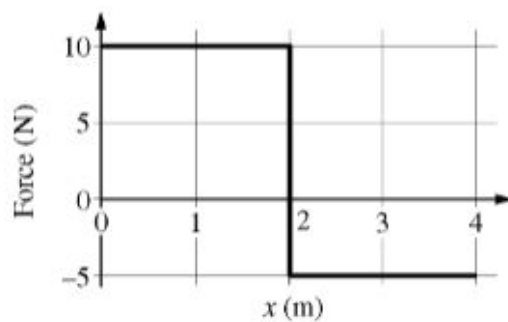


WARMUPS

CONSERVATION OF MECHANICAL
ENERGY IN CLOSED SYSTEMS LAB
DESIGN

A 120 g mass is pushed 50 cm from equilibrium on a horizontal spring as shown below. It reaches the equilibrium position with a speed of 950 cm/s. What is the spring constant, k , of the spring?





A cart is moving on a level track in the positive x -direction. A force acting parallel to the x -axis is exerted on the cart. The graph above shows the net force exerted on the cart as a function of displacement. As the cart travels from $x = 0$ m to $x = 4$ m, what is the net change in the kinetic energy of the cart?

(A) An increase of 20 J

(A)

(B) An increase of 10 J

(B)

(C) A decrease of 20 J

(C)

(D) A decrease of 10 J

(D)

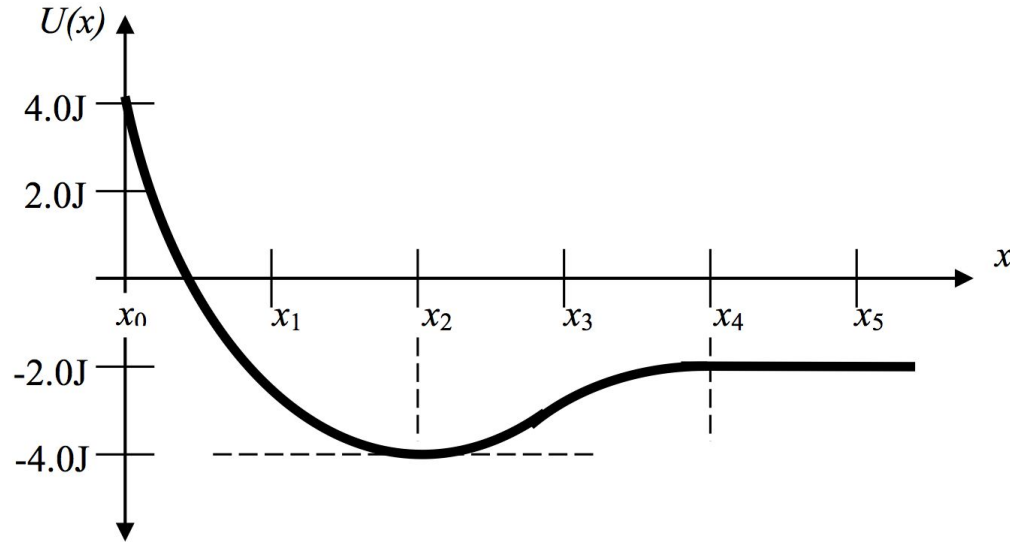
AGENDA FOR 01-20-2022/01-21 -2022



WARMUPS

CONSERVATION OF MECHANICAL
ENERGY IN CLOSED SYSTEMS QUIZ

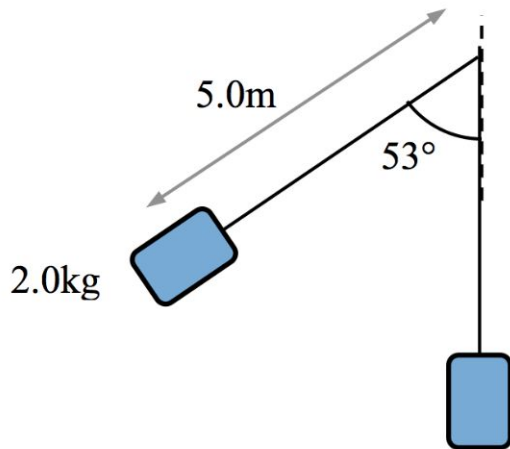
Question:



The potential energy function $U(x)$ is associated with a conservative force F and described by the graph given here. If a particle being acted upon by this force has a kinetic energy of 1.0 J at position x_0 , what is the particle's kinetic energy at position x_4 ?

- a. 6.0 J
- b. 7.0 J
- c. 2.0 J
- d. -2.0 J
- e. -7.0 J

Question:



A mass of 2.0 kg is attached to the end of a light cord to make a pendulum 5.0 meters in length. The mass is raised to an angle of 53° relative to the vertical, as shown, and released. The speed of the mass at the bottom of its swing is:

- a. 60 m/s
- b. 7.7 m/s
- c. 40 m/s
- d. 6.3 m/s
- e. 10 m/s

A 2 kg block collides with a massless spring of spring constant 100 N/m attached to a wall. The speed of the block was observed to be 1.4 m/s at the moment of collision.

The acceleration of gravity is 9.8 m/s^2 .

How far does the spring compress if the surface on which the mass moves is frictionless?

Answer in units of cm. Answer in units of cm.

A block of mass 2 kg is kept at rest as it compresses a horizontal massless spring ($k = 100 \text{ N/m}$) by 10 cm. As the block is released, it travels 0.35 m on a rough horizontal surface before stopping.

The acceleration of gravity is 9.8 m/s^2 .

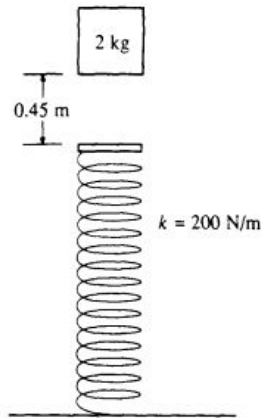
Calculate the coefficient of kinetic friction between surface and block.

Question:

From the top of a tall cliff of height y , a soccer ball is kicked horizontally so that it leaves the cliff with a velocity v . Assuming air friction is negligible, the speed of the ball just before it hits the ground is:

- a. $2gy$
- b. $\sqrt{2gy}$
- c. $v^2 + 2gy$
- d. $\sqrt{v^2 + 2gy}$
- e. $\sqrt{v^2 - 2gy}$

Try the first 2



A 2-kilogram block is dropped from a height of 0.45 meter above an uncompressed spring, as shown above. The spring has an elastic constant of 200 newtons per meter and negligible mass. The block strikes the end of the spring and sticks to it.

- Determine the speed of the block at the instant it hits the end of the spring.
- Determine the period of the simple harmonic motion that ensues.
- Determine the distance that the spring is compressed at the instant the speed of the block is maximum.

AGENDA FOR 01-24-2022/01-25 -2022



WARMUP

QUIZ

WORK

Energy conservation

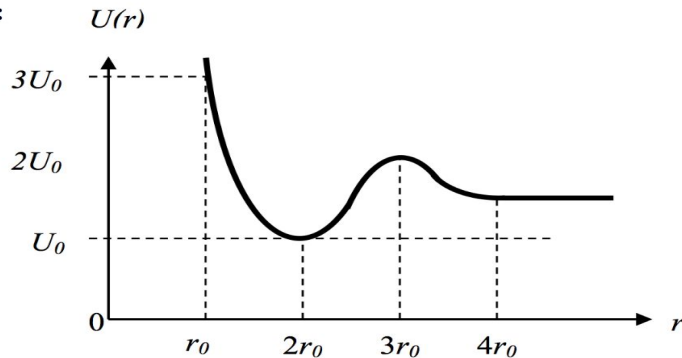
A 300 g ball is thrown upward at a speed of 40 m/s. Neglect air resistance.

1. Use energy conservation to determine the maximum height of the ball.
2. Use energy conservation to determine the speed of the ball at 60 m above the ground.

Energy conservation

A 500 g object is given an initial speed of 10 m/s up a 30 degree frictionless incline. How far along the incline does the object travel. Neglect air resistance.

Question:



The graph above represents the potential energy U as a function of position r for a particle of mass m . If the particle is released from rest at position r_0 , what will its speed be at position $3r_0$?

a. $\sqrt{\frac{8U_0}{m}}$

b. $\sqrt{\frac{4U_0}{m}}$

c. $\sqrt{\frac{2U_0}{m}}$

d. $\sqrt{\frac{6U_0}{m}}$

e. The particle will never reach position $3r_0$.

AGENDA FOR 01-24-2022/01-25 -2022



WARMUP

WORK

WORK-KINETIC ENERGY THEOREM

Question:

A moon of mass m orbits a planet of mass M in a perfectly circular orbit of radius R , with a force of gravitational attraction between the two bodies of F_g . How much Work is done on the moon by the planet during a single orbit of the moon ?

- a. 0
- b. $F_g R$
- c. $F_g M m R$
- d. $F_g 2\pi R$
- e. $F_g \pi R^2$

AGENDA FOR 01-28-2022/01-31 -2022



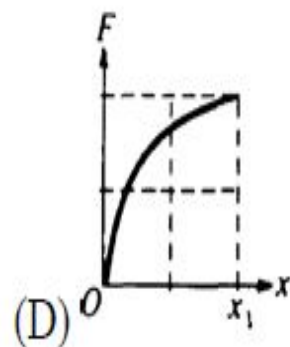
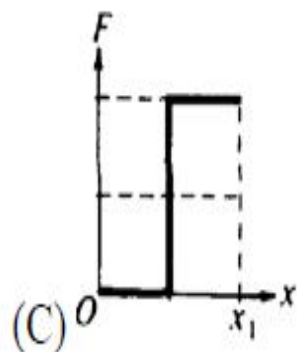
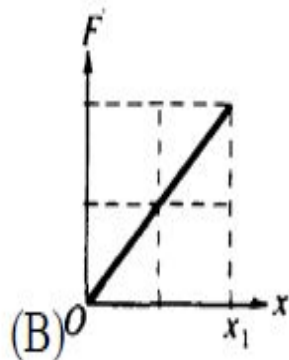
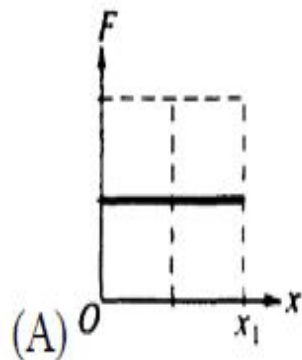
WARMUP

WORK

WORK-KINETIC ENERGY THEOREM

Practice

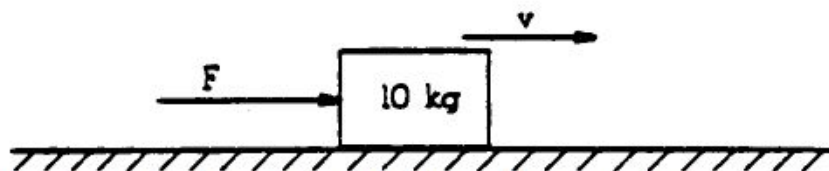
The following graphs, all drawn to the same scale, represent the net force F as a function of displacement x for an object that moves along a straight line. Which graph represents the force that will cause the greatest change in the kinetic energy of the object from $x = 0$ to $x = x_1$?



Question:

A 10-meter long, vertical cannon is used to accelerate a 1.0-kg ball straight up into the air. A constant force of 13.2-Newtons is used to accelerate the bowling ball up the length of the cannon. What is the ball's approximate velocity as it leaves the cannon (assuming no energy loss to friction)?

- a. 29 m/s
- b. 16 m/s
- c. 14 m/s
- d. 9 m/s
- e. 8 m/s



1981B1. A 10-kilogram block is pushed along a rough horizontal surface by a constant horizontal force F as shown above. At time $t = 0$, the velocity v of the block is 6.0 meters per second in the same direction as the force. The coefficient of sliding friction is 0.2. Assume $g = 10$ meters per second squared.

- a. Calculate the force F necessary to keep the velocity constant.

The force is now changed to a larger constant value F' . The block accelerates so that its kinetic energy increases by 60 joules while it slides a distance of 4.0 meters.

- b. Calculate the force F' .
- c. Calculate the acceleration of the block.

AGENDA FOR 0-2-07-2022/02-08 -2022



WARMUP

POWER

WORK- ENERGY QUIZ

Question:

A bucket of water with a total weight of 50 Newtons is lifted at constant velocity up a 10 meter deep well. If it takes 20 seconds to raise the bucket this distance, the Power required to lift the bucket is:

- a. 25 W
- b. 25 J
- c. 2.5 J
- d. 500 J
- e. 500 W

AGENDA FOR 0-2-15-2022/02-16- 2022



WARMUP

WEP TEST RESULTS

THE WRITE STUFF

WARMUP 2-15-22

Convert the following:

- 1) 8.5 kg/cubic meter to g/cubic meter
- 2) 29.4 square meters to square centimeters
- 3) 650 mi/hr squared to m/s squared

AGENDA FOR 0-2-17-2022/02-18- 2022



WARMUP

MOMENTUM AND IMPULSE

OPEN AND CLOSED SYSTEMS

WARMUP

Convert

- 1) 3900 mN to N
- 2) 27570 N/cm to N/m
- 3) 4.5×10^9 mm squared to m squared
- 4) 8000 μ J to J ($\mu = 1 \times 10^{-6}$)

AGENDA FOR 02-16A/20B

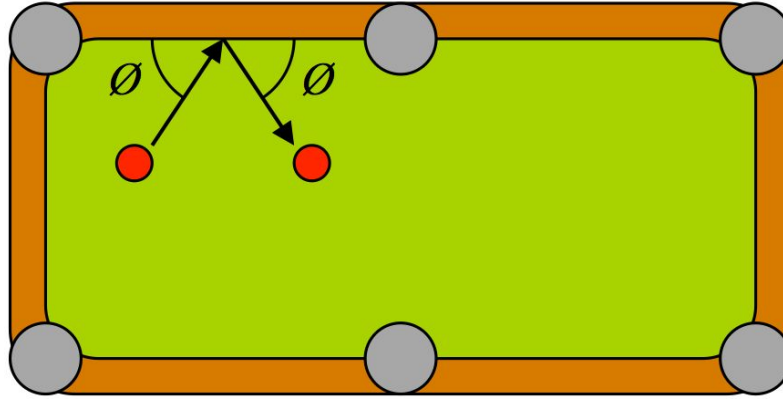


DAILY QUIZ

WARMUP

MOMENTUM AND IMPULSE

Question:



View looking
down on pool
table

A billiard ball hits the side of a pool table at an angle θ as shown in the top view above, and bounces away at the same angle, and with the same speed. Which vector indicates the direction of the net change in momentum of the billiard ball?

a.



b.



c.



d.



e.



AGENDA FOR

0-2-22-2022/02-23

-2022

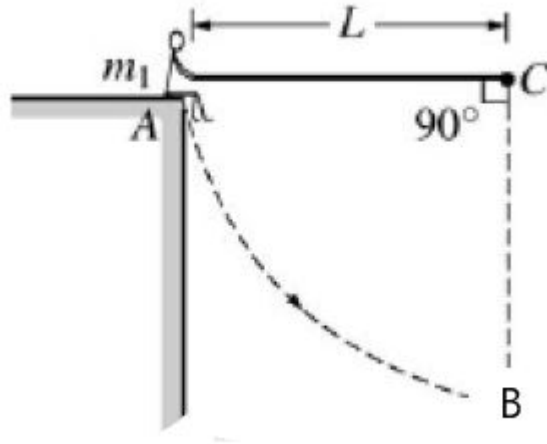


WARMUP

MOMENTUM AND IMPULSE QUIZ

WARMUP

The 60 kg person in the figure below starts from rest at point A and swings to point B. If the string is 5 m long and the person is moving at 6 m/s at point B. Calculate the work done by air resistance in moving from point A to point B.



WARMUP

CONVERT THE FOLLOWING

1A) 750 000 cm squared to m squared

3A) 2.8 m squared to mm squared

WARMUP

CONVERT THE FOLLOWING

2B) 750 000 ms squared to s squared

4B) 2.8 km squared to m squared

AGENDA FOR 02-21A/22B



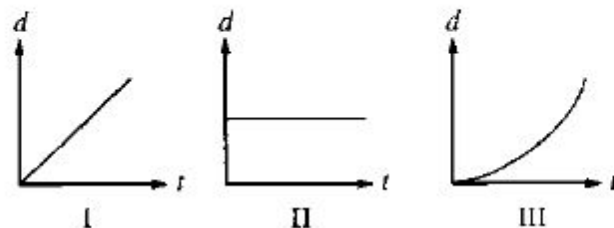
WARMUP

UNITS POP QUIZ

MOMENTUM AND IMPULSE

COLLISIONS

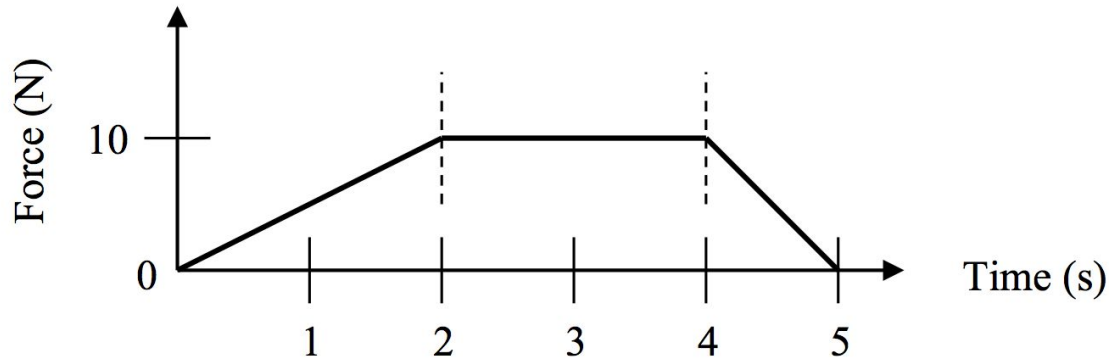
Three objects can only move along a straight, level path. The graphs below show the position d of each of the objects plotted as a function of time t .



The magnitude of the momentum of the object is increasing in which of the cases?

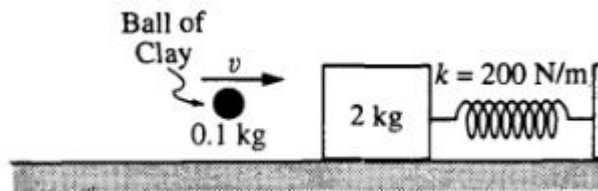
(A) II only (B) III only (C) I and II only (D) I and III only

Question:



An object of mass $m = 2.0\text{kg}$ experiences a force in Newtons according to the *Force vs. time* graph shown here. For the time interval shown, what is the total change in momentum of the object?

- a. $35 \text{ kg}\cdot\text{m/s}$
- b. $70 \text{ kg}\cdot\text{m/s}$.
- c. $-35 \text{ kg}\cdot\text{m/s}$
- d. $-70 \text{ kg}\cdot\text{m/s}$
- e. none of these.



A 2-kilogram block is attached to an ideal spring (for which $k = 200 \text{ N/m}$) and initially at rest on a horizontal frictionless surface, as shown in the diagram above.

In an initial experiment, a 100-gram (0.1 kg) ball of clay is thrown at the 2-kilogram block. The clay is moving horizontally with speed v when it hits and sticks to the block. The spring is attached to a wall as shown. As a result, the spring compresses a maximum distance of 0.4 meters.

- Calculate the energy stored in the spring at maximum compression.
- Calculate the speed of the clay ball and 2-kilogram block immediately after the clay sticks to the block but before the spring compresses significantly.
- Calculate the initial speed v of the clay.

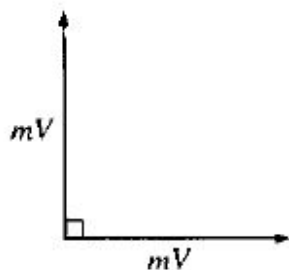
AGENDA FOR 0-2-22-2022/02-23 -2022



WARMUP

COLLISIONS

POP CHARTS



A stationary object explodes, breaking into three pieces of masses m , m , and $3m$. The two pieces of mass m move off at right angles to each other with the same magnitude of momentum mV , as shown in the diagram above. What are the magnitude and direction of the velocity of the piece having mass $3m$?

Magnitude

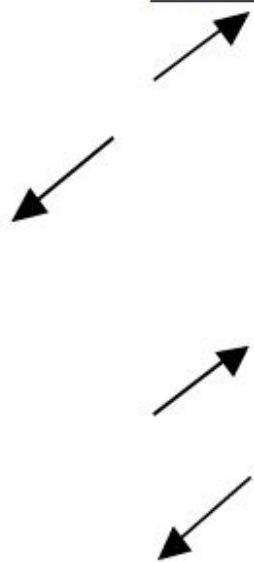
Direction

(A) $\frac{V}{\sqrt{2}}$

(B) $\frac{V}{\sqrt{2}}$

(C) $\frac{\sqrt{2}V}{3}$

(D) $\frac{\sqrt{2}V}{3}$



AGENDA FOR 0-3-9-2022/03-10- 2022



THE WRITE STUFF
EXTRA CREDIT

THE WRITE STUFF

Describe the similarities and differences in a perfectly elastic, perfectly inelastic, and recoil collisions. Give examples. Also mention how you can tell if a collision is elastic or inelastic if the objects collide and separate after the collision.

AGENDA FOR 3-

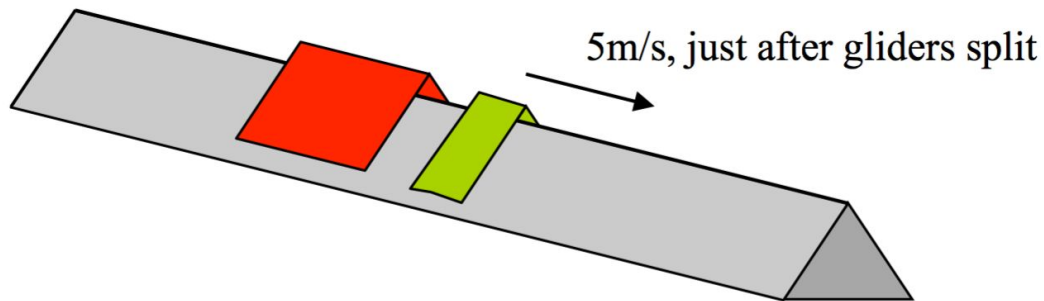


WARMUP

QUIZ QUIZ TRADE FOR WEP AND MOM

ROTATIONAL KINEMATICS

Question:



An air track glider of mass M is built, consisting of two smaller connected gliders with a small explosive charge located between them. The glider is traveling along a frictionless rail at 2 m/s to the right when the charge is detonated, causing the smaller glider with mass $\frac{1}{4}M$, to move off to the right at 5 m/s. What is the final velocity of the second small glider?

- a. 4 m/s to the left
- b. 2 m/s to the left
- c. 1 m/s to the left
- d. 0 m/s
- e. 1 m/s to the right

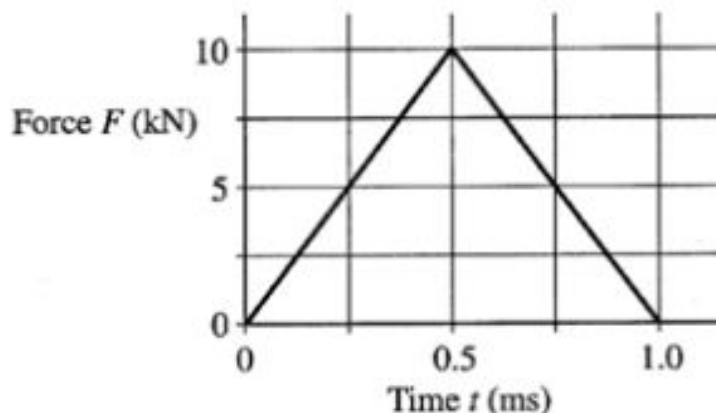


Two balls are on a frictionless horizontal tabletop. Ball X initially moves at 10 meters per second, as shown in Figure I above. It then collides elastically with identical ball Y which is initially at rest. After the collision, ball X moves at 6 meters per second along a path at 53° to its original direction, as shown in Figure II above. Which of the following diagrams best represents the motion of ball Y after the collision?

- (A) (B) (C) (D)



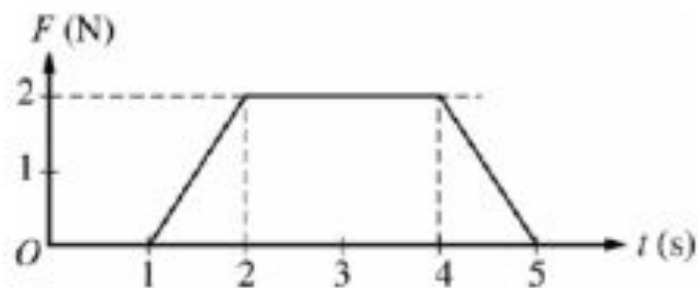
2002B1B. A 2.0 kg frictionless cart is moving at a constant speed of 3.0 m/s to the right on a horizontal surface, as shown above, when it collides with a second cart of undetermined mass m that is initially at rest. The force F of the collision as a function of time t is shown in the graph below, where $t = 0$ is the instant of initial contact. As a result of the collision, the second cart acquires a speed of 1.6 m/s to the right. Assume that friction is negligible before, during, and after the collision.



- Calculate the magnitude and direction of the velocity of the 2.0 kg cart after the collision.
- Calculate the mass m of the second cart.

A 2 kg object initially moving with a constant velocity is subjected to a force of magnitude F in the direction of motion. A graph of F as a function of time t is shown. What is the increase, if any, in the velocity of the object during the time the force is applied?

- (A) 0 m/s (B) 3.0 m/s (C) 4.0 m/s (D) 6.0 m/s

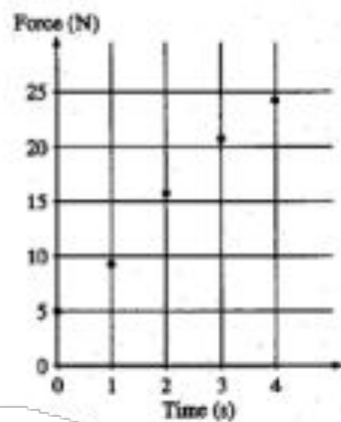


A mass m has speed v . It then collides with a stationary object of mass $2m$. If both objects stick together in a perfectly inelastic collision, what is the final speed of the newly formed object?

- (A) $v/3$ (B) $v/2$ (C) $2v/3$ (D) $3v/2$

A student obtains data on the magnitude of force applied to an object as a function of time and displays the data on the graph shown. The increase in the momentum of the object between $t=0$ s and $t=4$ s is most nearly

- (A) 40 N·s
- (B) 50 N·s
- (C) 60 N·s
- (D) 80 N·s





WARMUP

ROTATIONAL MOTION QUIZ REVIEW

TORQUE

Uniform Acceleration 101079

A car accelerates uniformly from rest and covers a distance of 57 m in 8.3 s.

If the *diameter* of a tire is 30 cm, find the angular acceleration of the wheel.

Answer in units of rad/s^2 .

Find the final angular velocity of one of the car's wheels.

Answer in units of rad/s .

Find total number of rotations of the wheel.



WARMUP

ROTATIONAL KINEMATICS QUIZ

TORQUE

A coin with a diameter of 2.37 cm is dropped onto a horizontal surface. The coin starts out with an initial angular speed of 18.9 rad/s and rolls in a straight line without slipping.

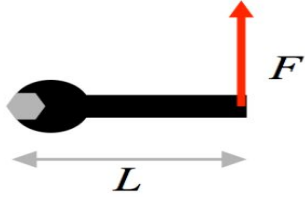
If the rotation slows with an angular deceleration of 1.22 rad/s^2 , how far does the coin roll before coming to rest?

Answer in units of m.

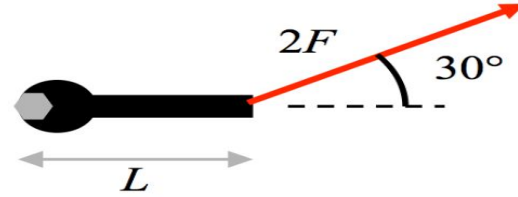
Question:

A series of wrenches of different lengths is used on a hexagonal bolt, as shown below. Which combination of wrench length and Force applies the greatest torque to the bolt?

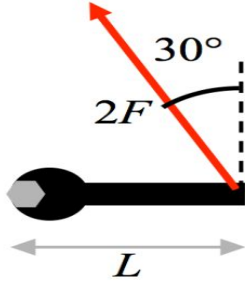
a.



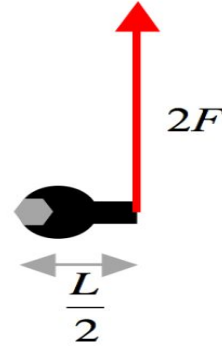
b.



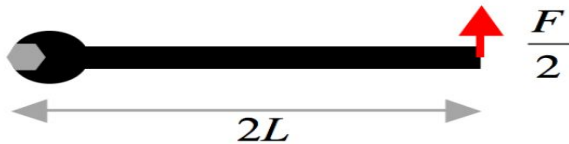
c.

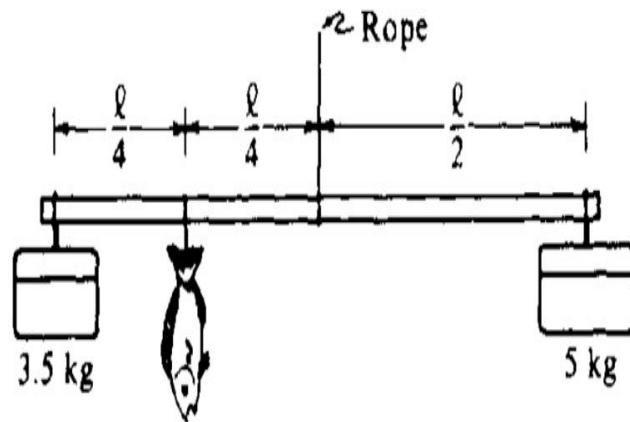


d.



e.

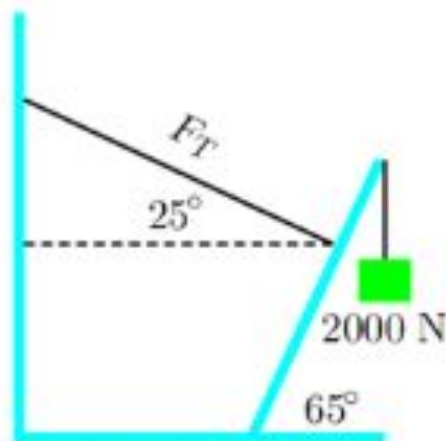




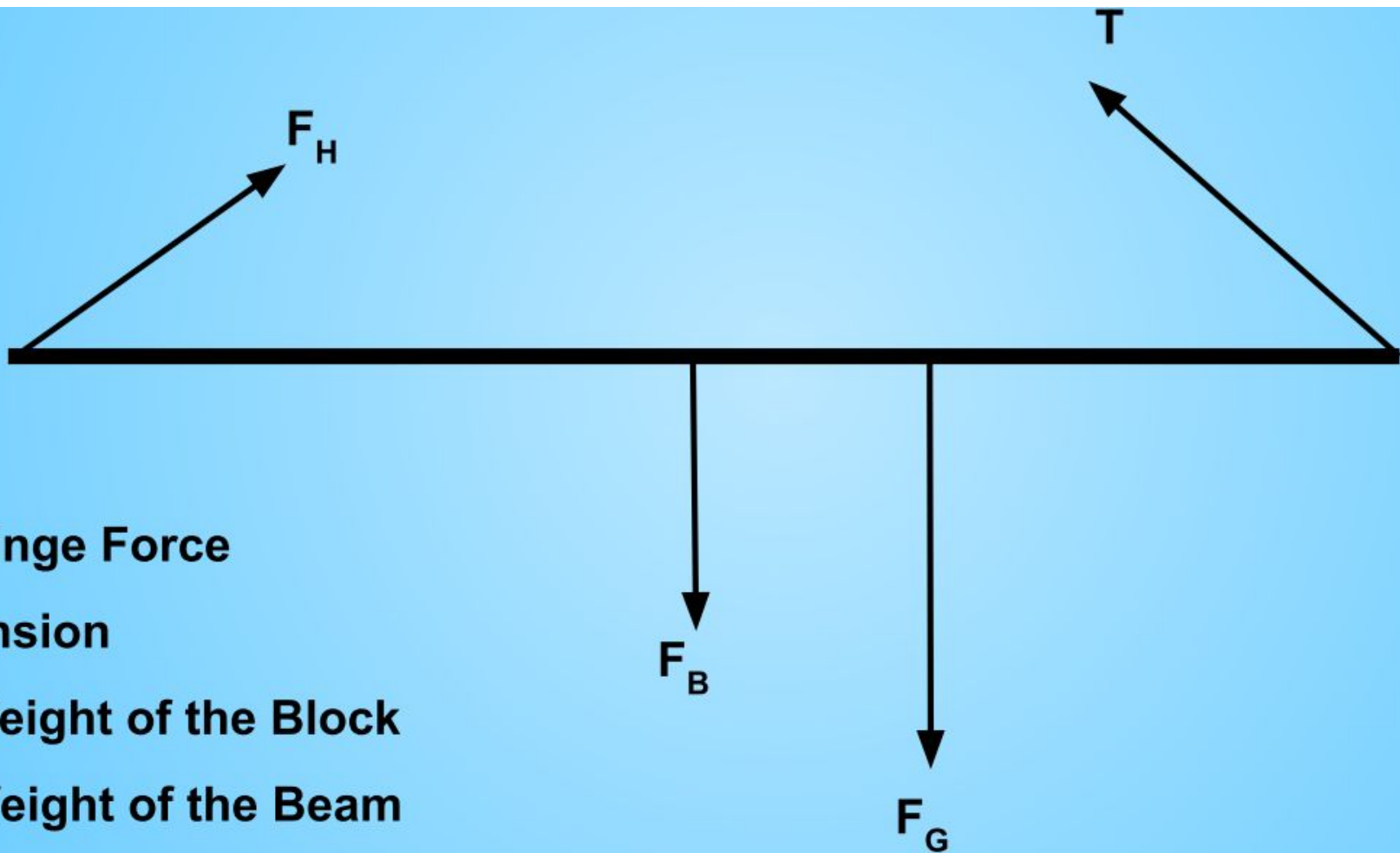
8. To weigh a fish, a person hangs a tackle box of mass 3.5 kilograms and a cooler of mass 5 kilograms from the ends of a uniform rigid pole that is suspended by a rope attached to its center. The system balances when the fish hangs at a point $1/4$ of the rod's length from the tackle box. What is the mass of the fish?

- (A) 2 kg (B) 6.5 kg (C) 1.5 kg (D) 6 kg (E) 3 kg

A 1200 N uniform boom of length ℓ is supported by a cable, as shown. The boom is pivoted at the bottom, the cable is attached a distance $\frac{3}{4} \ell$ from the pivot, and a 2000 N weight hangs from the boom's top.



Find the force F_T applied by the supporting cable.

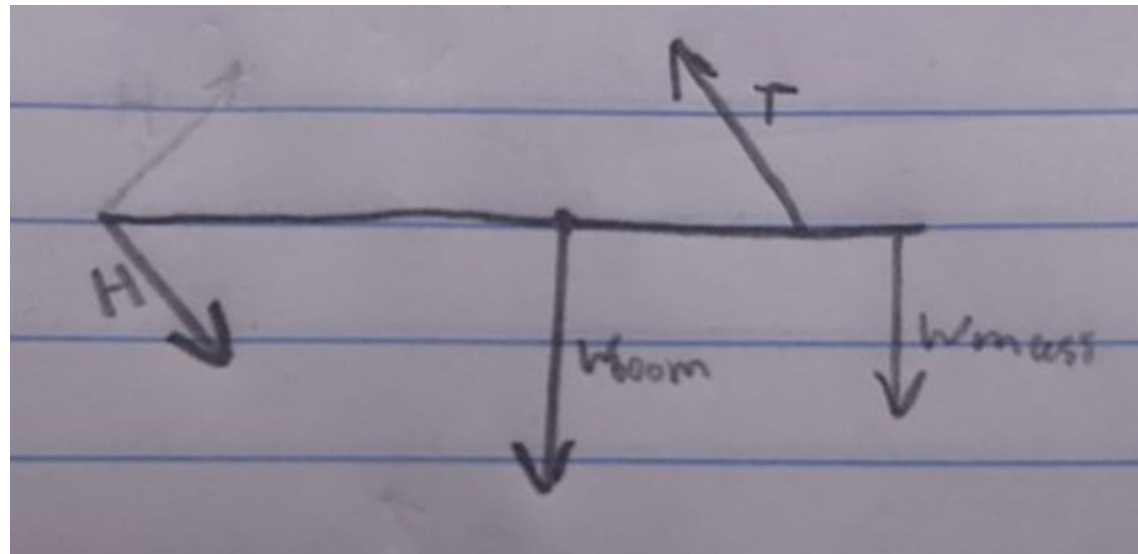


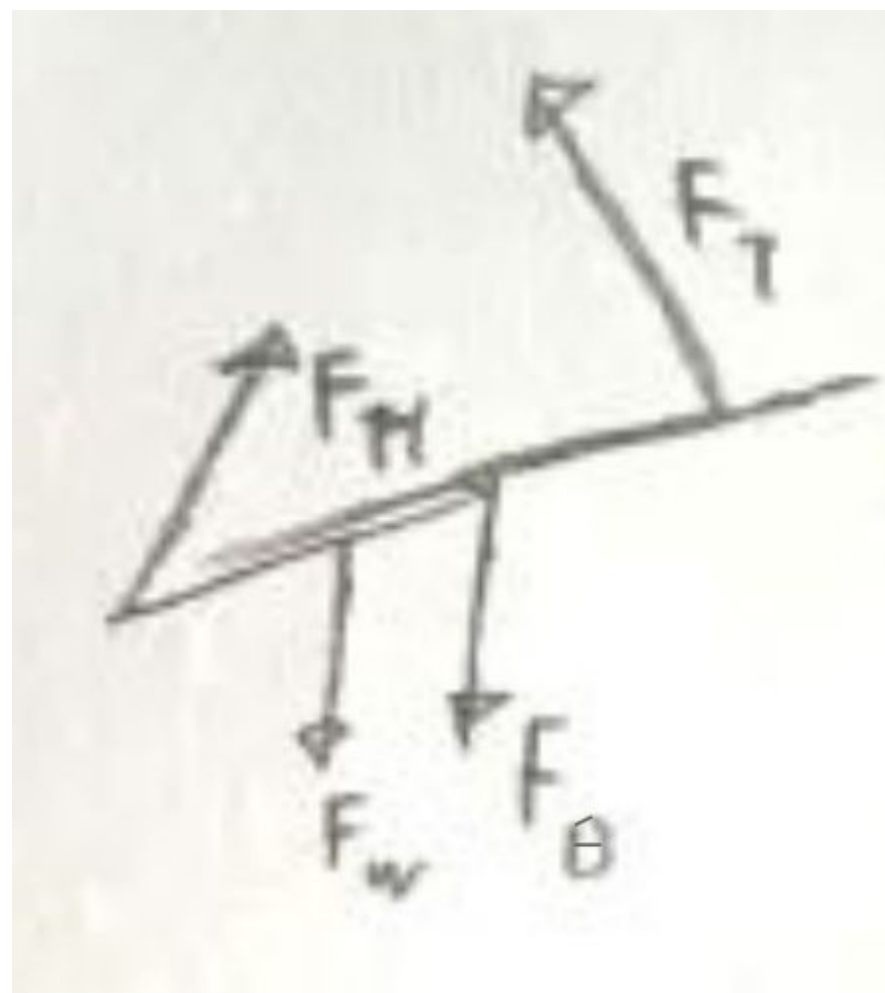
F_H = Hinge Force

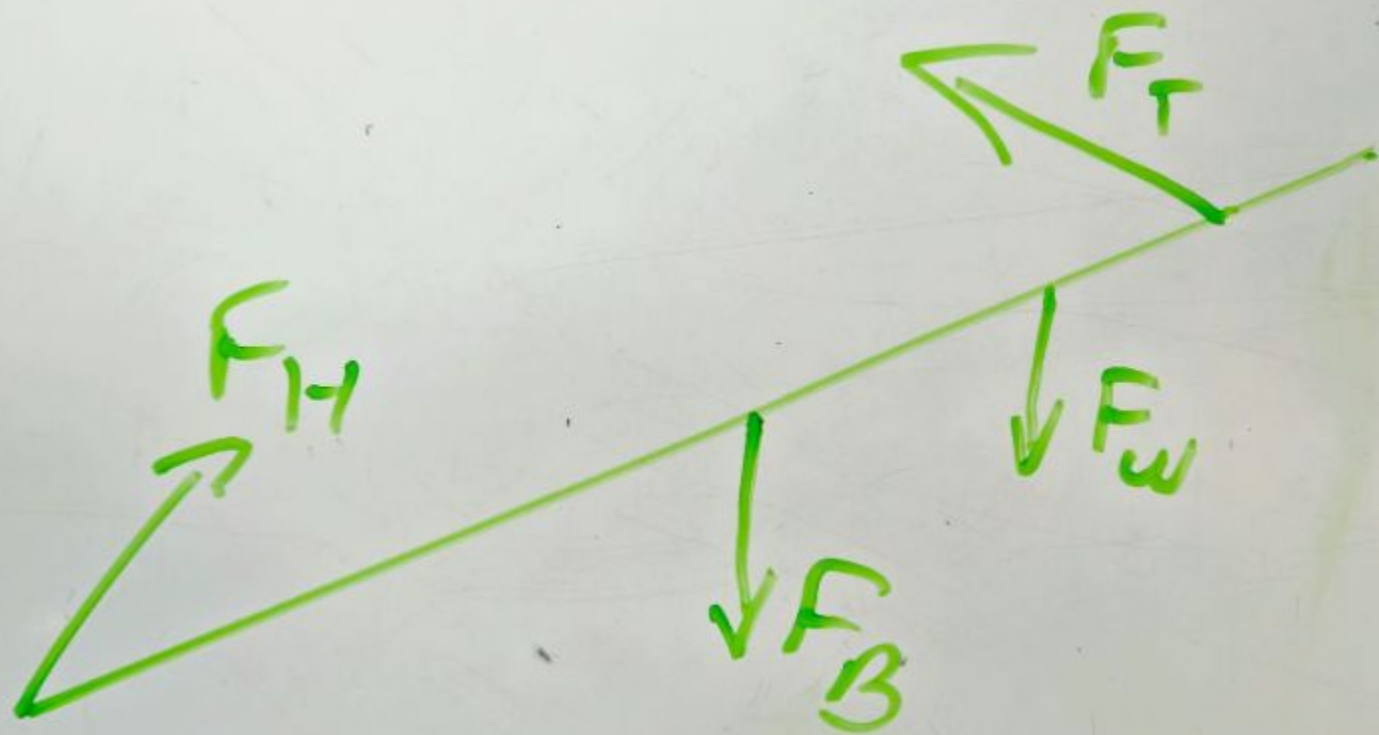
T = Tension

F_B = Weight of the Block

F_G = Weight of the Beam



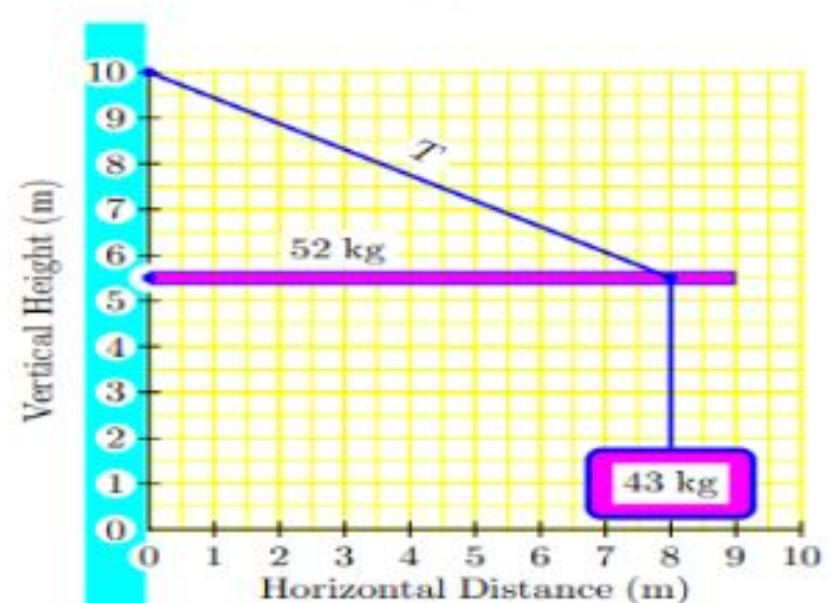


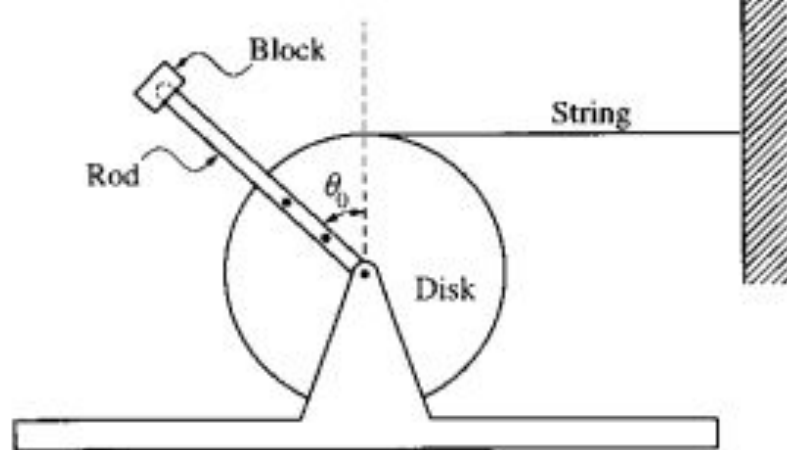


STATIC EQUILIBRIUM

A weight (with a mass of 43 kg) is suspended from a point near the right-hand end of a uniform boom with a mass of 52 kg. To support the uniform boom a cable runs from this same point to a wall (left side of the figure) and by pivot on the same wall at an elevation of 5.5 m.

Calculate the tension, T , in the cable.





1999M3 As shown above, a uniform disk is mounted to an axle and is free to rotate without friction. A thin uniform rod is rigidly attached to the disk so that it will rotate with the disk. A block is attached to the end of the rod. Properties of the disk, rod, and block are as follows.

Disk: mass = $3m$, radius = R ,

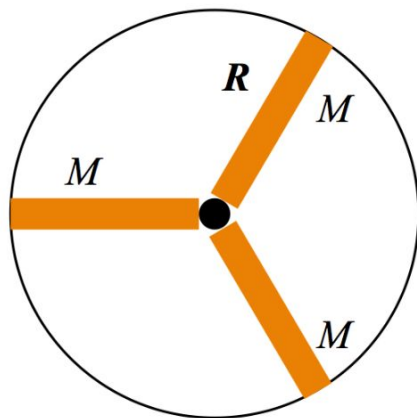
Rod: mass = m , length = $2R$,

Block: mass = $2m$

The system is held in equilibrium with the rod at an angle θ_0 to the vertical, as shown above, by a horizontal string of negligible mass with one end attached to the disk and the other to a wall. Express your answers to the following in terms of m , R , θ_0 , and g .

- Determine the tension in the string.

Question:



A wheel consists of a three uniform spokes, with length R and mass M , mounted 120 degrees apart on a horizontal frictionless axle and connected by a rim of negligible mass. Consider the counterclockwise direction to be positive. When the spokes are oriented as shown in the diagram above, the net Torque on the wheel due to the weight of the spokes is

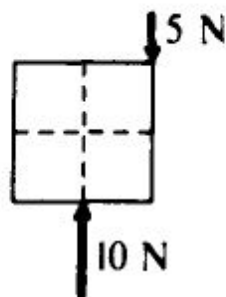
- a. $+RMg$
- b. $-2RMg$
- c. $\frac{R}{2}Mg(\sqrt{3}-1)$
- d. $\frac{R}{2}Mg(1-\sqrt{3})$
- e. 0

Question:

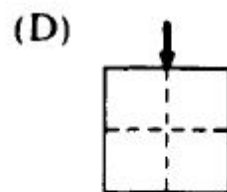
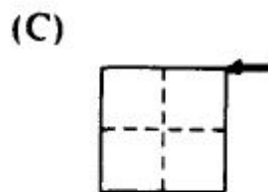
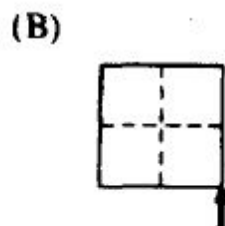
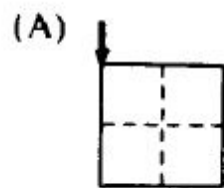
A turntable begins to rotate from rest with a constant angular acceleration. If a point on the edge of the turntable travels 9 radians during a 3.0 second time period, the angular acceleration of the turntable is

- a. $2.0s^{-2}$
- b. $3.0s^{-2}$
- c. $4.0s^{-2}$
- d. $6.0s^{-2}$
- e. $9.0s^{-2}$

- **A uniform meterstick is supported by a string placed at the 40 cm mark. A 50 g mass is placed at the 80 cm mark and a 100 g mass is placed at the 0 cm mark. What is the mass of the meterstick?**



A square piece of plywood on a horizontal tabletop is subjected to the two horizontal forces shown above. Where should a third force of magnitude 5 newtons be applied to put the piece of plywood into equilibrium?





ROTATION UNIT REVIEWS

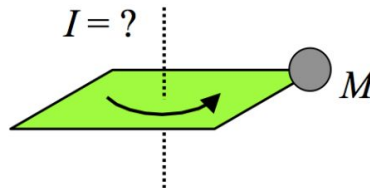
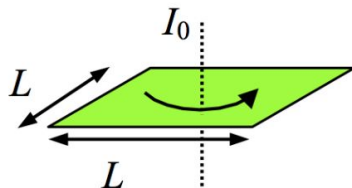


WARMUP

MOMENT OF INERTIA AND ROTATIONAL KINETIC
ENERGY

ENERGY CONSERVATION

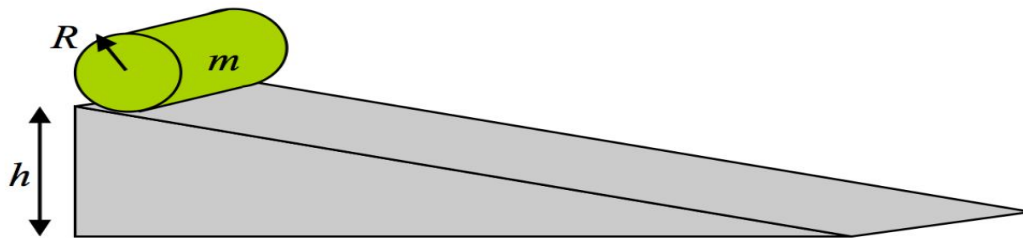
Question:



A square of mass M and sides of length L has a moment of inertia I_0 when rotated about an axis perpendicular to its surface and passing through its center, as shown. Now a lump of clay, also of mass M is attached to one corner of the square as shown. What is the new moment of inertia of the masses about the same axis of rotation?

- a. $I_0 + \frac{ML^2}{4}$
- b. $I_0 + \frac{ML^2}{2}$
- c. $I_0 + \frac{\sqrt{2}ML^2}{2}$
- d. $I_0 + 2ML^2$
- e. $I_0 + ML^2$

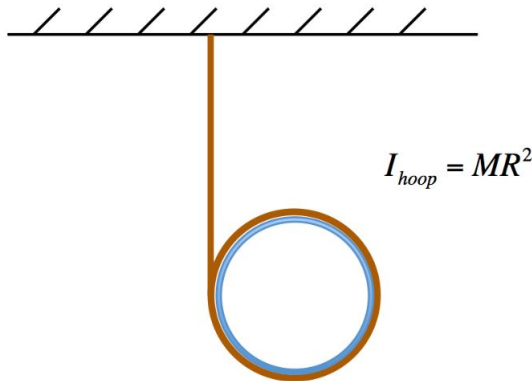
Question:



A cylinder of mass m and radius R has a moment of inertia of $\frac{1}{2}mR^2$. The cylinder is released from rest at a height h on an inclined plane, and rolls down the plane without slipping. What is the velocity v of the cylinder when it reaches the bottom of the incline?

- a. $\sqrt{\frac{4}{3}gh}$
- b. $\sqrt{\frac{1}{2}gh}$
- c. $\sqrt{\frac{3}{4}gh}$
- d. $\sqrt{\frac{3}{4}mgh}$
- e. $\sqrt{gh}$

Question:



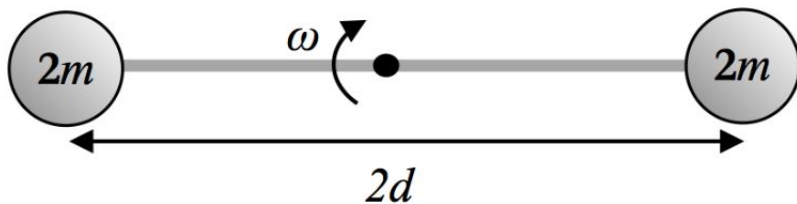
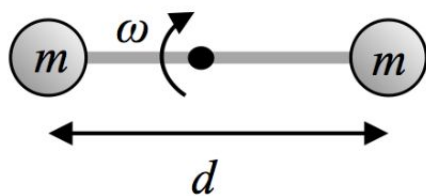
A light cord is wrapped around a hoop of mass M , radius R , and moment of inertia $I_{hoop} = MR^2$. The cord is attached to the ceiling, and the hoop released from rest so that it falls toward the floor, the cord unwinding as it falls. The tension in the cord as the hoop falls is

- a. Mg
- b. $\frac{Mg}{R}$
- c. $\frac{Mg}{2}$
- d. $\frac{Mg}{2R}$
- e. $\frac{g}{MR^2}$



ROTATION UNIT FRQ TEST

Question:



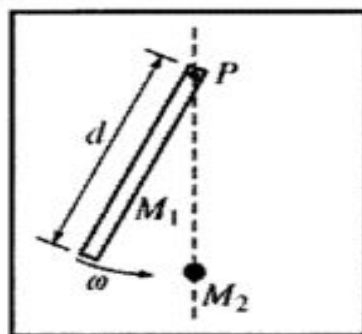
A dumbbell consists of two masses m connected by a rigid rod of negligible mass and length d . A physics student takes the dumbbell and rotates it about its center of mass with an angular velocity ω , giving it an angular momentum L_1 . The student then takes a second dumbbell, with masses $2m$ and length $2d$, and rotates them with the same angular velocity ω . What is the angular momentum L_2 of this second dumbbell?

- a. $2 L_1$
- b. $4 L_1$
- c. $6 L_1$
- d. $8 L_1$
- e. $16 L_1$

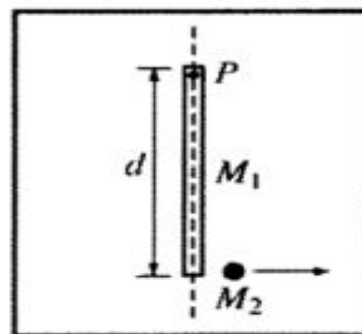
Question:

A long, thin, rod of with moment of inertia $I=2 \text{ kg}\cdot\text{m}^2$ is free to rotate about an axis passing through the midpoint of the rod. The rod begins rotating from rest at time $t=0$ seconds, accelerating constantly so that it has a rotational velocity of $4\pi \text{ rad/s}$ after rotating through two complete revolutions. What is the rod's angular momentum at this point?

- a. $2\pi \text{ kg}\cdot\text{m}^2/\text{s}$
- b. $8\pi \text{ kg}\cdot\text{m}^2/\text{s}$
- c. $8 \text{ kg}\cdot\text{m}^2/\text{s}$
- d. $4\pi \text{ kg}\cdot\text{m}^2$
- e. $4 \text{ kg}\cdot\text{m}^2$



Before Collision



After Collision

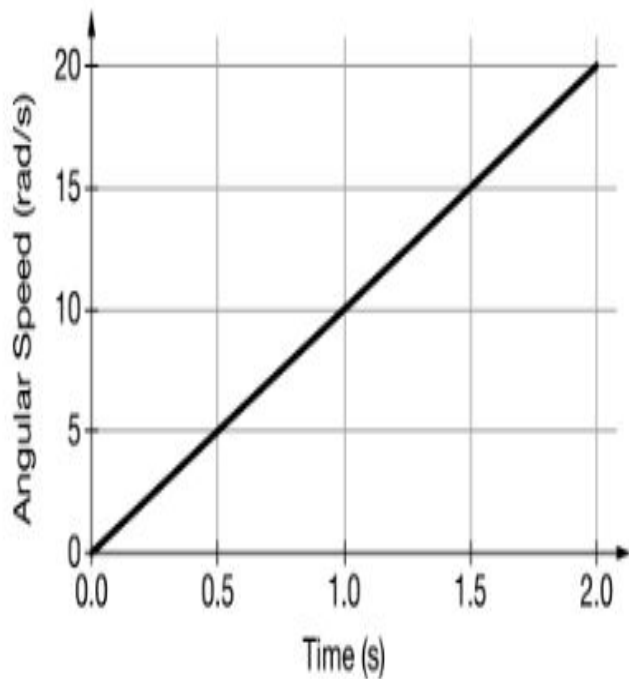
TOP VIEWS

2005 M3

A system consists of a ball of mass M_2 and a uniform rod of mass M_1 and length d . The rod is attached to a horizontal frictionless table by a pivot at point P and initially rotates at an angular speed ω , as shown above left. The rotational inertia of the rod about point P is $\frac{1}{3} M_1 d^2$. The rod strikes the ball, which is initially at rest.

As a result of this collision, the rod is stopped and the ball moves in the direction shown above right. Express all answers in terms of M_1 , M_2 , ω , d , and fundamental constants.

- Derive an expression for the angular momentum of the rod about point P before the collision.
- Derive an expression for the speed v of the ball after the collision.



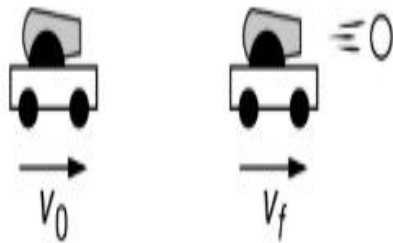
An object rotates with an angular speed that varies with time, as shown in the graph. How can the graph be used to determine the magnitude of the angular acceleration α of the object? Justify your selection.

A Subtract the greatest value of the angular speed from the smallest value of the angular speed, because $\alpha = \Delta\omega$.

B Determine the slope of the line from 0 s to 2 s, because the slope represents $\frac{\Delta\omega}{\Delta t}$.

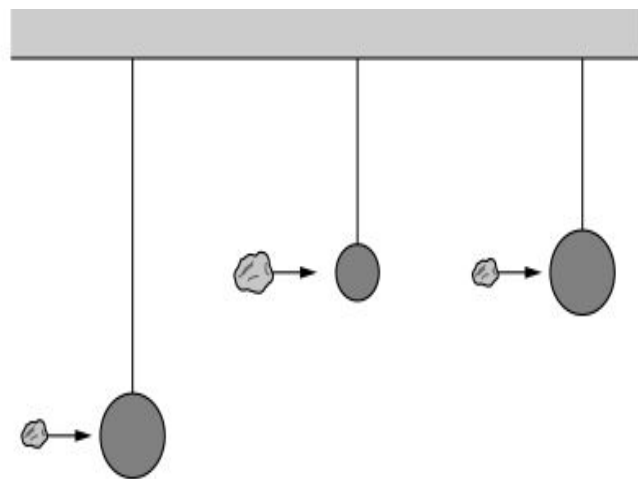
C Determine the area bounded by the line and the horizontal axis from 0 s to 2 s, because $\alpha = \frac{1}{2}\omega\Delta t$.

D The angular acceleration cannot be determined without knowing the rotational inertia of the object.



A cart with a toy projectile launcher attached to its top travels forward at a constant speed v_0 . The launcher fires a solid sphere forward at a speed much greater than that of the cart-launcher system. The cart's speed after firing the dart is $v_f < v_0$. Frictional forces are considered to be negligible. Which of the following scenarios best represents the rotational analog to this situation?

- (A)** A gaseous planet rotates with a large rotational velocity, causing the middle of the planet's equator to bulge outward.
- (B)** A spinning platform falls onto a stationary platform. Both platforms rotate together, but at a smaller common rotational speed than that of the spinning platform immediately before the collision.
- (C)** A child on the edge of a large rotating disk throws a ball in the same direction as the child's tangential velocity.
- (D)** A figure skater spins with the skater's hands at the skater's waist. When the skater extends the skater's arms outward, the skater's rotational velocity decreases.



Scenario 1:	Scenario 2:	Scenario 3:
$m_{\text{sphere}} = 500 \text{ g}$	$m_{\text{sphere}} = 250 \text{ g}$	$m_{\text{sphere}} = 500 \text{ g}$
$\ell = 0.10 \text{ m}$	$\ell = 0.05 \text{ m}$	$\ell = 0.05 \text{ m}$
$m_{\text{clay}} = 40 \text{ g}$	$m_{\text{clay}} = 60 \text{ g}$	$m_{\text{clay}} = 40 \text{ g}$

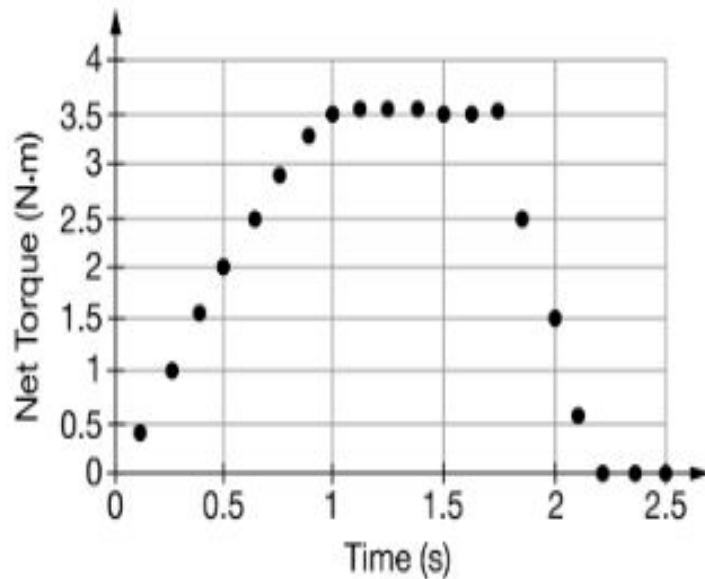
A lump of clay of mass m_{clay} with speed $v_{\text{clay}} = 8 \text{ m/s}$ travels toward various spheres that are suspended from the ceiling by lightweight strings of different lengths, as shown in the figure. For the three scenarios, the clay collides with the suspended sphere and sticks to it. Which of the following correctly relates the angular momentum L of the clay-bob system immediately after the collision for each scenario, where the angular momentum is taken about the point where the string is attached to the ceiling?

(A) $L_2 > L_1 = L_3$

(B) $L_1 = L_2 = L_3$

(C) $L_1 > L_2 = L_3$

(D) $L_1 > L_2 > L_3$



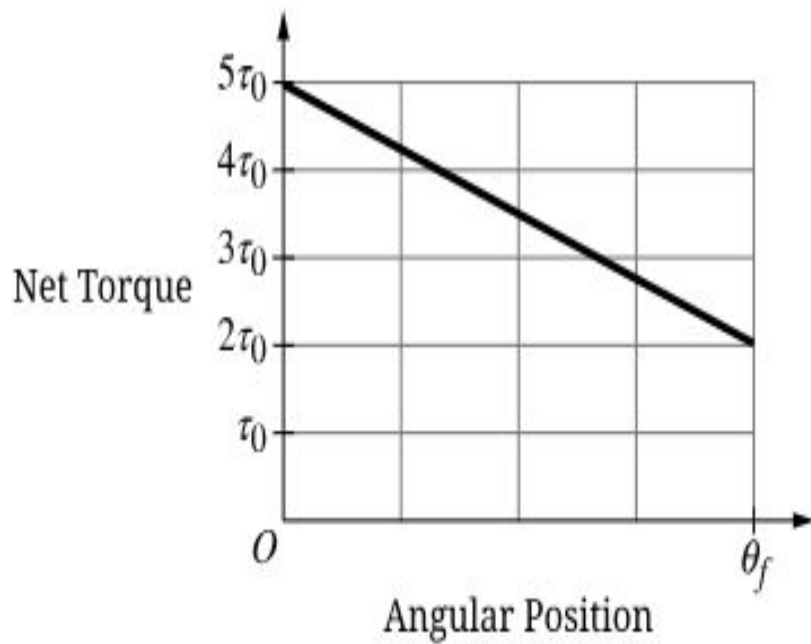
(A) 0.00 s

(B) 1.00 s

(C) 1.75 s

(D) 2.50 s

A horizontal disk of radius 0.2 m and mass 0.3 kg is mounted on a central vertical axle so that a student can study the relationship between net torque and change in angular momentum of the disk. In the experiment, the student uses a force probe to collect data pertaining to the net torque exerted on the edge of the disk as a function of time, as shown in the graph. The disk is initially at rest. At what instant in time does the disk have the greatest angular momentum?



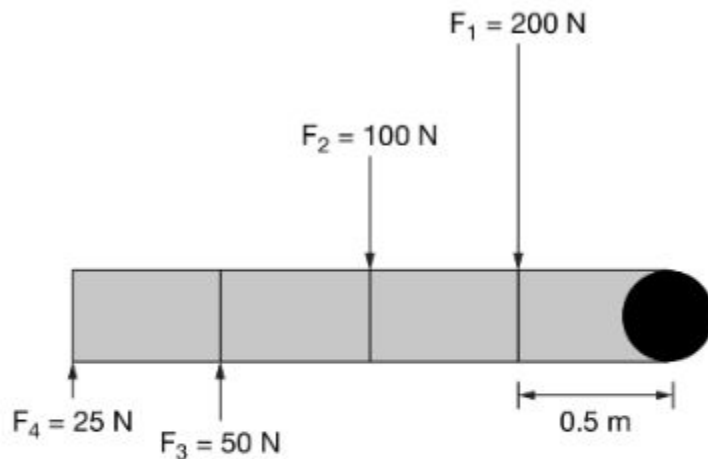
(A) $-\frac{3\tau_0}{\theta_f}$

(B) $2\tau_0\theta_f$

(C) $\frac{7}{2}\tau_0\theta_f$

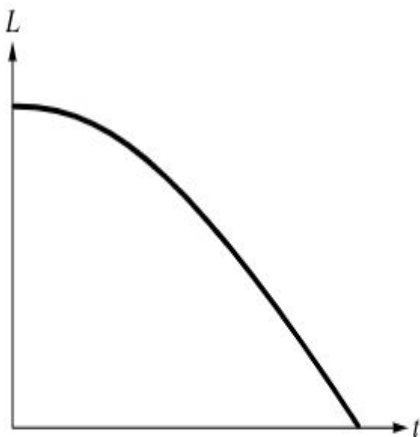
(D) $5\tau_0\theta_f$

The net torque exerted on a rigid system as a function of the system's angular position is represented in the graph shown. Which of the following expressions is equal to the work done on the system?



The figure above represents a stick of uniform density that is attached to a pivot at the right end and has equally spaced marks along its length. Any one or a combination of the four forces shown can be exerted on the stick as indicated.

All four forces are exerted on the stick that is initially at rest. What is the angular momentum of the stick after 2.0 s ?



The graph shows the angular momentum L of a rigid system as a function of time t . Which of the following statements about the torque exerted on the system is supported by the graph?

- (A) The torque decreases in magnitude and is in the same direction as the angular momentum.
- (B) The torque decreases in magnitude and is in the opposite direction of the angular momentum.
- (C) The torque increases in magnitude and is in the same direction as the angular momentum.
- (D) The torque increases in magnitude and is in the opposite direction of the angular momentum.

Question:

A simple pendulum has a period T on the earth. What is the period T' of this same pendulum on the moon, where the acceleration due to gravity is $1/6$ that of the earth?

a. $\frac{T}{\sqrt{6}}$

b. $\frac{T}{6}$

c. $\sqrt{6} T$

d. $6T$

e. $36T$

Question:

An enormous pendulum-driven clock, located on the earth, is set into motion by releasing its 10-meter long simple pendulum from a maximum angle of less than 10° relative to the vertical. At what approximate time t will the pendulum have fallen to a perfectly vertical orientation?

- a. $\frac{\pi}{10}$ seconds
- b. $\frac{\pi}{5}$ seconds
- c. $\frac{\pi}{4}$ seconds
- d. $\frac{\pi}{2}$ seconds
- e. π seconds

Question:

A particle moves constantly in a circle centered at the origin with a period of 4.0 seconds. If its position at time $t = 0$ seconds is (2,0) meters, two possible equations describing the particle's x - and y -components are:

a. $x = 2\cos\left(\frac{\pi}{2}t\right)$ $y = 2\sin\left(\frac{\pi}{2}t\right)$

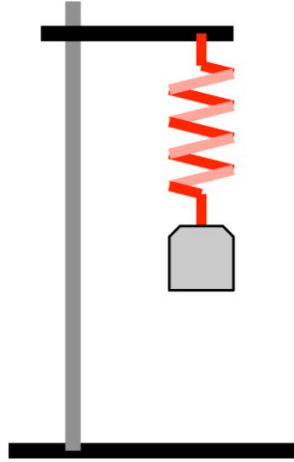
b. $x = 2\cos\left(\frac{2}{\pi}t\right)$ $y = 2\sin\left(\frac{2}{\pi}t\right)$

c. $x = 2\sin\left(\frac{\pi}{2}t\right)$ $y = 2\cos\left(\frac{\pi}{2}t\right)$

d. $x = 2\sin\left(\frac{\pi}{2}\right)$ $y = 2\cos\left(\frac{\pi}{2}\right)$

e. $x = 2\pi\cos(2t)$ $y = 2\pi\sin(2t)$

Question:



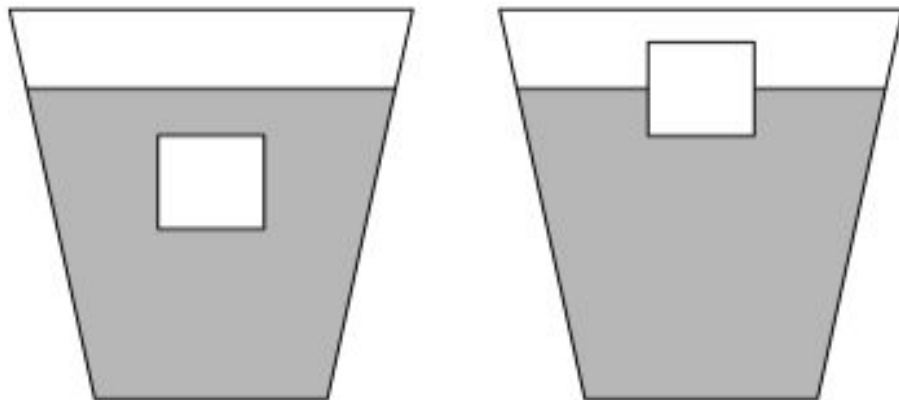
One end of a light spring with spring constant 10 N/m is attached to a vertical support, while a mass is attached to the other end, as shown. The mass is pulled down and released, and exhibits simple harmonic motion with a period of 0.2π . The mass is

- a. 0.1 kg
- b. 0.25 kg
- c. 0.4 kg
- d. 0.04 kg
- e. 0.025 kg

The density of copper is 8960 kg/m^3 . The mass of a copper sphere with a radius of 0.03 m is most nearly

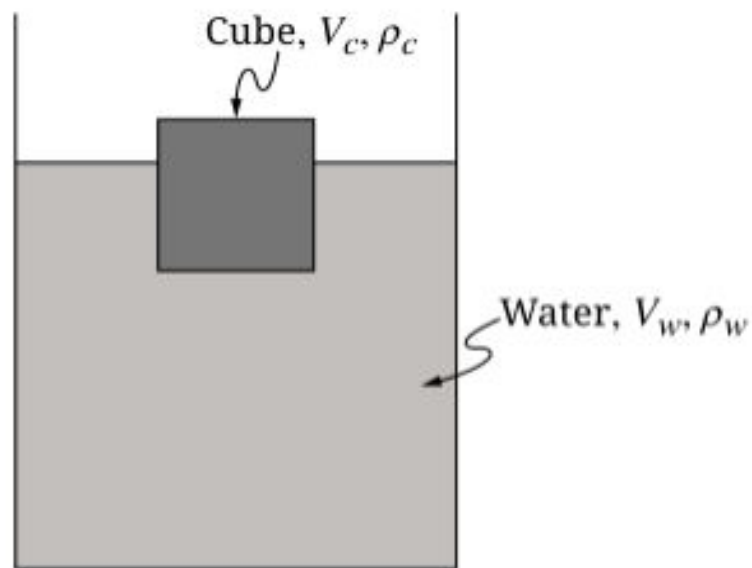
- (A) $1.26 \times 10^{-8} \text{ kg}$
- (B) 1.01 kg
- (C) 25.3 kg
- (D) $7.92 \times 10^7 \text{ kg}$

A block of unknown mass is fully submerged in Fluid 1 of density ρ_1 . The block remains in place when it is released. The same block is then placed in Fluid 2 of density ρ_2 . The block floats in Fluid 2 with half of the volume of the block submerged in the fluid, as shown in the figure.



Which of the following correctly indicates the ratio $\frac{\rho_1}{\rho_2}$?

- (A) $\frac{1}{4}$
- (B) $\frac{1}{2}$
- (C) 1
- (D) 2



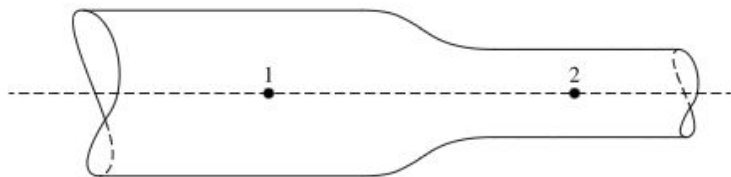
A large container holds a volume V_w of water that has a density ρ_w . A cube with volume V_c and density ρ_c floats partially submerged in the water. Which of the following expressions is equal to the buoyant force on the cube?

- (A) $\rho_c V_c g$
- (B) $\rho_w V_c g$
- (C) $\rho_c V_w g$
- (D) $\rho_w V_w g$

A small sphere with volume $V = 3.7 \times 10^{-4} \text{ m}^3$ is released from rest at the surface of a large pool of water which is 3.0 m deep. The density of water is $\rho_{\text{water}} = 1000 \text{ kg/m}^3$. The sphere reaches the bottom of the pool in 1.5 s. Drag force from the water is negligible.

What is the approximate mass of the sphere?

- (A) 0.29 kg
- (B) 0.50 kg
- (C) 1.39 kg
- (D) 3.75 kg



Water flows to the right through the pipe shown in the figure. Points **1** and **2** lie on the same horizontal line, but the diameter of the pipe at Point **2** is smaller than the diameter of the pipe at Point **1**. Which of the following quantities is the same at both Points **1** and **2**?

(A)

The speed of the water

~~**(A)**~~

(B)

The pressure of the water

~~**(B)**~~

(C)

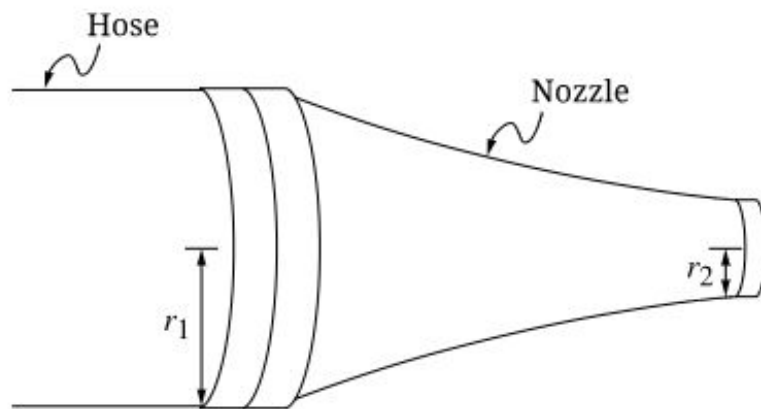
The flow rate of the water

~~**(C)**~~

(D)

The cross-sectional area of the pipe

~~**(D)**~~



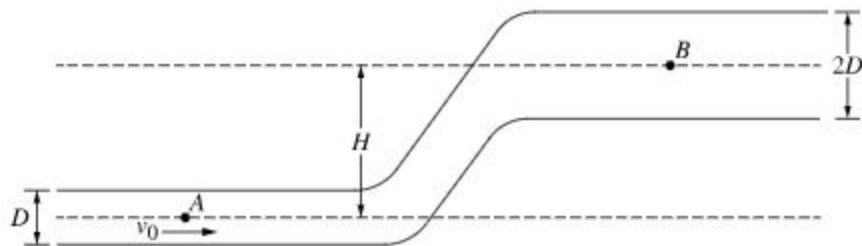
A hose is connected to a nozzle, as shown in the figure. Both the hose and the nozzle have circular cross-sections. Water flows in the hose with a speed of 0.3 m/s . The radius r_1 of the hose is 0.02 m and the radius r_2 of the nozzle at the point where the water exits is 0.002 m . What is the speed of the water as it exits the nozzle?

(A) 0.003 m/s

(B) 0.3 m/s

(C) 3 m/s

(D) 30 m/s



Water of density ρ flows through the section of circular pipe shown in the figure. At Point **A**, where the diameter of the pipe is D , the water has a pressure P_0 and velocity v_0 . Point **B** is located a vertical distance H above Point **A** in a section of the pipe that has diameter $2D$.

Which of the following expressions is equal to the pressure of the water at Point **B**?

(A) $P_0 - \rho g H$

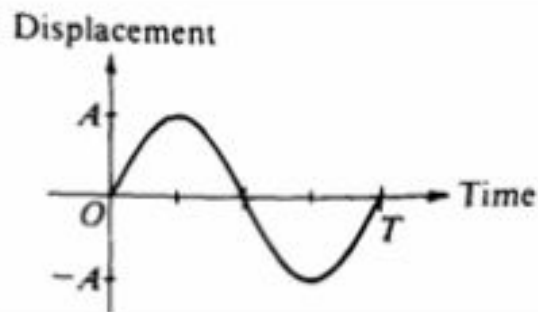
(C) $P_0 - \rho g H + \frac{15}{32} \rho v_0^2$

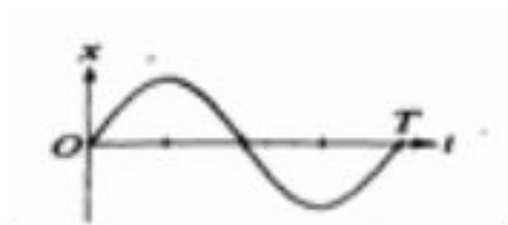
(B) $P_0 - \rho g H + \frac{3}{8} \rho v_0^2$

(D) $2P_0$

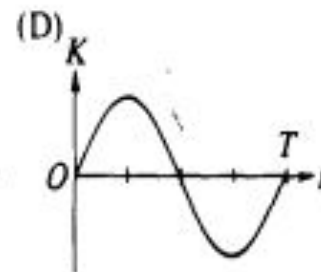
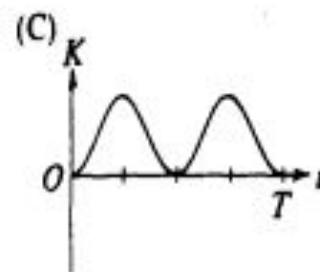
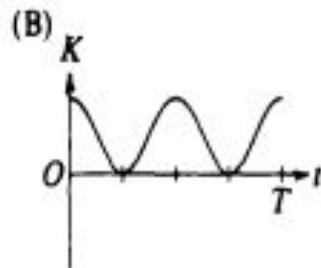
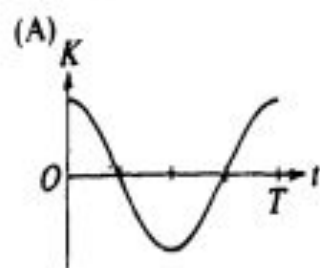
An object is attached to a spring and oscillates with amplitude A and period T , as represented on the graph. The nature of the velocity v and acceleration a of the object at time $T/4$ is best represented by which of the following?

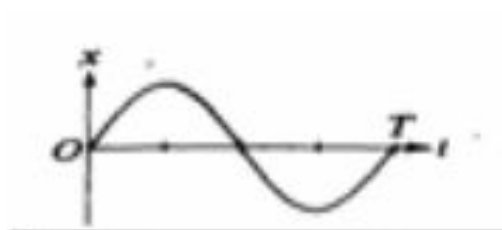
- (A) $v > 0$, $a > 0$ (B) $v > 0$, $a < 0$
(C) $v > 0$, $a = 0$ (D) $v = 0$, $a < 0$





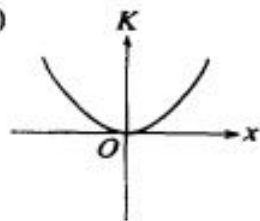
Which of the following graphs shows the kinetic energy K of the particle as a function of time t for one cycle of motion?



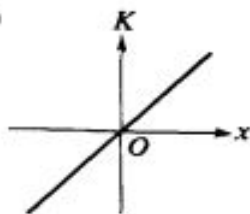


Which of the following graphs shows the kinetic energy K of the particle as a function of its displacement x ?

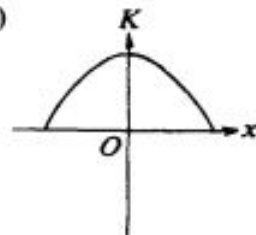
(A)



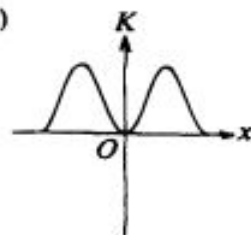
(B)



(C)



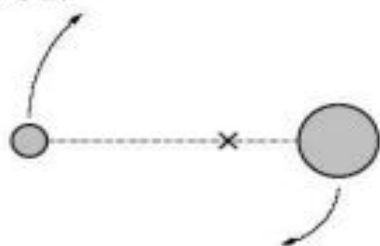
(D)



A satellite of mass m and speed v moves in a stable, circular orbit around a planet of mass M . What is the radius of the satellite's orbit?

- (A) $\frac{Gv}{mM}$ (B) $\frac{GM}{v^2}$ (C) $\frac{GmM}{v}$ (D) $\frac{GmM}{v^2}$

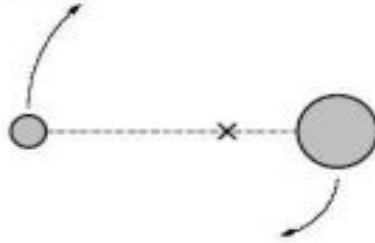
Two stars orbit their common center of mass as shown in the diagram below. The masses of the two stars are $3M$ and M . The distance between the stars is d .



What is the value of the gravitational potential energy of the two star system?

- (A) $-\frac{GM^2}{d}$ (B) $\frac{3GM^2}{d}$ (C) $-\frac{GM^2}{d^2}$ (D) $-\frac{3GM^2}{d}$

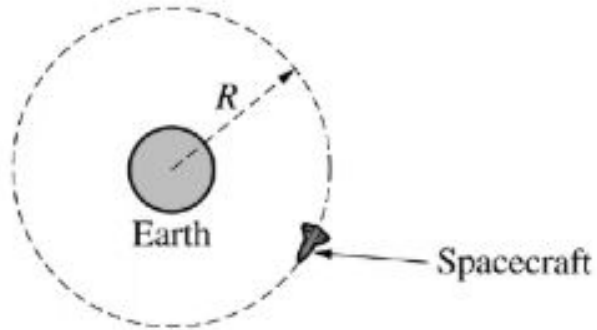
Two stars orbit their common center of mass as shown in the diagram below. The masses of the two stars are $3M$ and M . The distance between the stars is d .



The mark, x , represents the center of mass of the two stars. What is this in terms of d .

- a) $d/2$ b) $5/6d$ c) $2/3d$ d) $3/4d$

GOALLESS GRAVITATION



Note: Figure not drawn to scale.

A spacecraft of mass m is in a clockwise circular orbit of radius R around Earth, as shown in the figure above.

The mass of Earth is M_E .